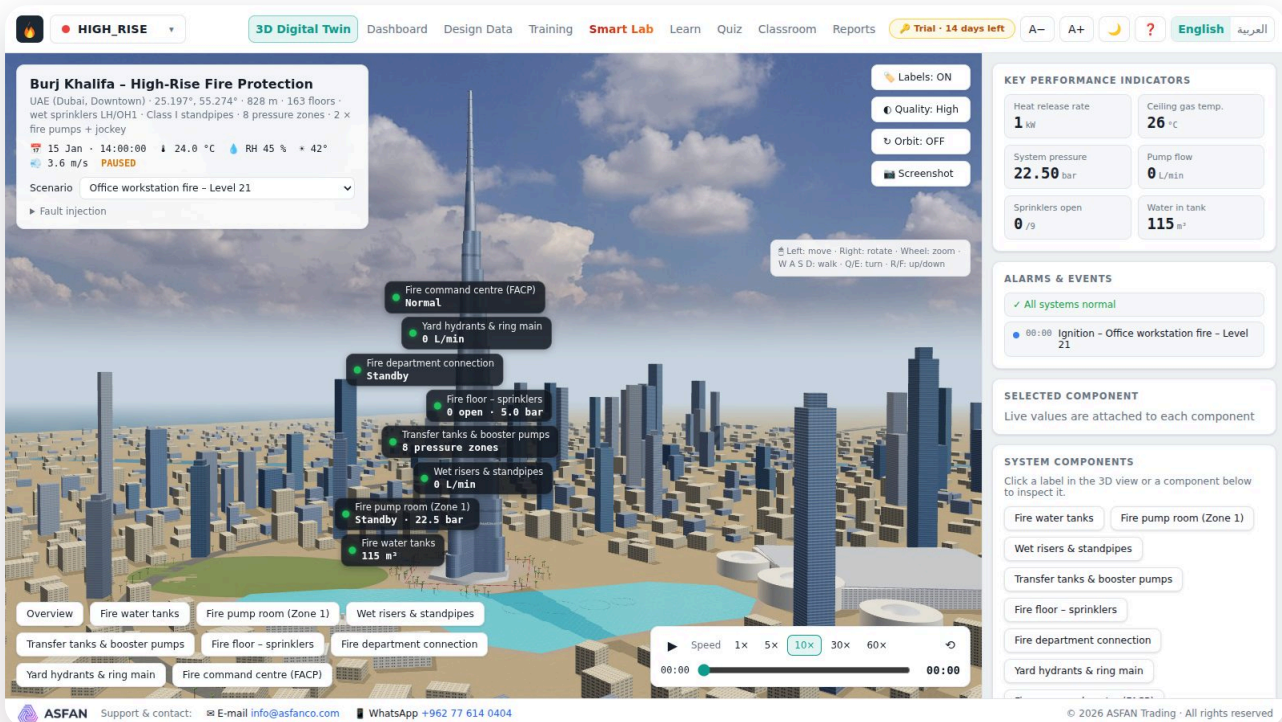




# Fire Protection Digital Twin

User Manual · Educational fire-fighting & smart fire-alarm simulator

Version 1.1 · Windows 10 / 11 (64-bit) · English



## ASFAN

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**Fire Protection Digital Twin** is an educational real-time simulator of fire-fighting and fire-alarm systems for universities, training centres and companies. Real sites (Burj Khalifa, Aqaba port, Amman data centre, oil tank farm, airport hangar) are modelled in 3D with their fire-protection systems designed to NFPA standards. A physics engine simulates fire growth, detection, sprinkler operation, pumps, water supply, clean agents and foam — and every value is shown live on the 3D model.

## What you can do

- Watch complete fire scenarios in 3D, with live pressures, flows, temperatures and alarms on every component.
- Inject faults (power failure, closed valve, jockey pump failure, leaks...) and diagnose them.
- Edit the design data and run the simulation on your own design, with NFPA compliance checks.
- Practise real procedures in ten interactive 3D training scenes, scored step by step.
- Learn smart (addressable) fire-alarm systems in the Smart Systems Lab: panel, cause & effect, BMS integration, commissioning, maintenance and incident command.
- Study lessons, take quizzes, run classroom challenges, print reports and certificates.

### Educational use

Building data are public information; the fire systems are NFPA-based reference designs for teaching. Do not use results for real projects without verification by a qualified engineer.

## 2 System requirements

---

|                  | Minimum                                                              | Recommended                        |
|------------------|----------------------------------------------------------------------|------------------------------------|
| Operating system | Windows 10 64-bit                                                    | Windows 11 64-bit                  |
| Processor        | Intel Core i5 (8th gen) / Ryzen 5                                    | Core i7 / Ryzen 7                  |
| Memory           | 8 GB RAM                                                             | 16 GB RAM                          |
| Graphics         | Integrated GPU with DirectX 11 / WebGL 2                             | Dedicated GPU (NVIDIA / AMD), 2 GB |
| Screen           | 1366 × 768                                                           | 1920 × 1080 or larger              |
| Disk space       | about 500 MB                                                         |                                    |
| Internet         | Not required to use the program; used only to check for new versions |                                    |

## 3 Installing the program

Download the latest version from the official download page [github.com/asfantrading-create/firetwin-releases](https://github.com/asfantrading-create/firetwin-releases) (or the link sent by ASFAN). Two files are available:

| File                                   | Use                                                                                                                       |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| FireProtectionDigitalTwin-Setup.exe    | <b>Installer (recommended).</b> Installs into Program Files, creates desktop and Start-menu shortcuts and an uninstaller. |
| FireProtectionDigitalTwin-Portable.exe | <b>Portable.</b> Runs directly without installation — e.g. from a USB stick or a lab PC without admin rights.             |

### Installation steps

- 1 Double-click `FireProtectionDigitalTwin-Setup.exe` (the file name may include the version number).
- 2 Windows may show a blue “**Windows protected your PC**” window (Microsoft SmartScreen). Click **More info**, then **Run anyway**. This appears because the program is new and is not yet signed with a paid code-signing certificate — it is safe to continue.
- 3 Choose the installer language (English / العربية) and accept the license agreement.
- 4 Choose **for all users** or **only for me**, and the installation folder (the default is fine).
- 5 Click **Install**, then **Finish**. The program starts automatically; a shortcut **Fire Protection Digital Twin** is placed on the desktop.

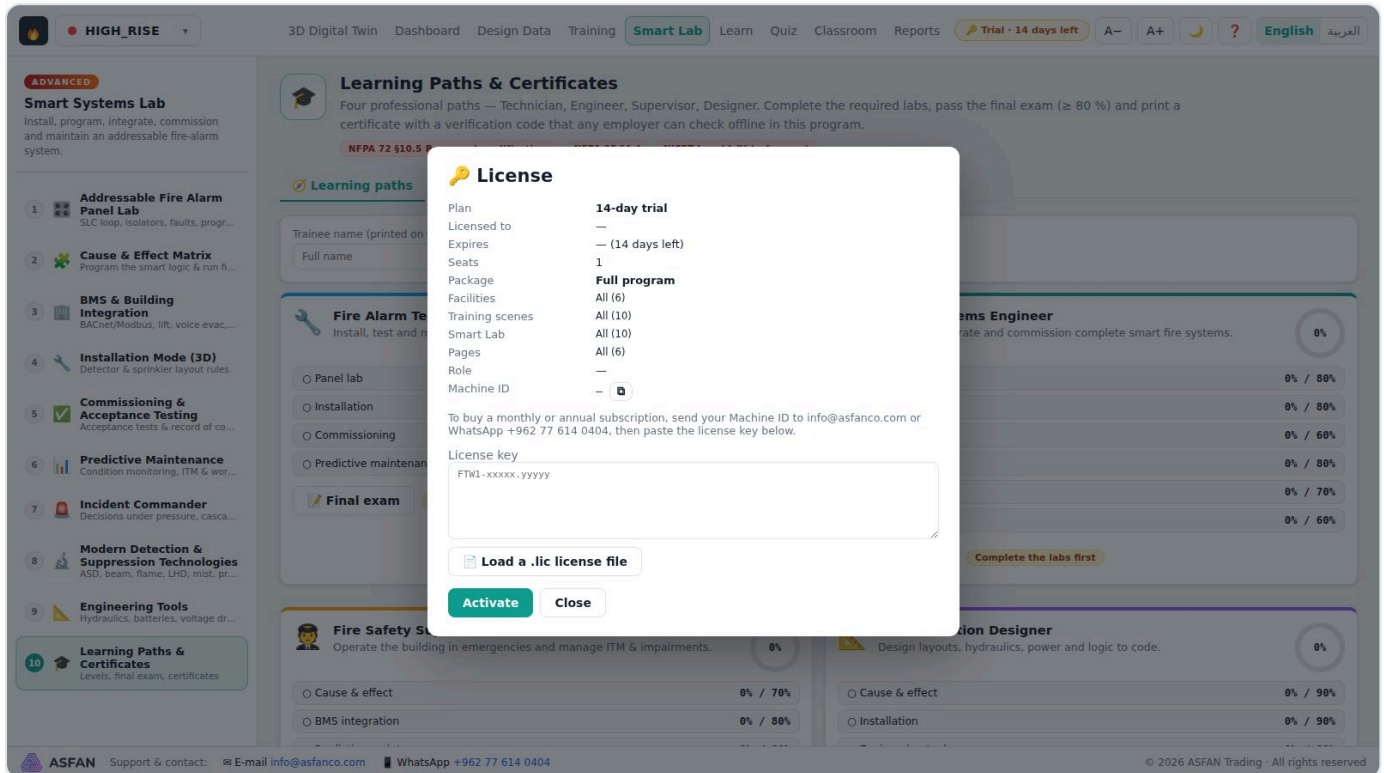
#### Antivirus

Some antivirus programs may scan the new file for a few seconds or ask for confirmation the first time. Allow the file — it is the official ASFAN installer. If a school network blocks unknown programs, ask the IT administrator to allow the installer or use the portable version.

### Updating and uninstalling

- **Update notice:** when the computer is online, the program checks for a new version at start-up (and from **Help → Check for updates**). A green bar appears at the top: **A new version is available** — click **What's new** or **Download**.
- **Update:** run the new Setup file — it replaces the old version. Your license, settings and results are kept.
- **Uninstall:** Windows Settings → Apps → Fire Protection Digital Twin → Uninstall.

On the first start the program runs as a **14-day full trial**. The license badge at the top (🔑) shows the remaining days.



The license window: plan, customer, expiry, package contents and the Machine ID

## Buying and activating a subscription

- 1 Click the 🔑 badge at the top of the window (or **Help** → **License...**).
- 2 Copy your **Machine ID** with the 📋 button and send it with your order to ASFAN (e-mail or WhatsApp) — only needed if your license should be bound to this computer.
- 3 ASFAN sends you a license file `.lic` (or a key starting with `FTW1-`).
- 4 In the license window click **Load a .lic license file** (or paste the key) and press **Activate**. The badge changes to **Licensed · Monthly / Annual**.

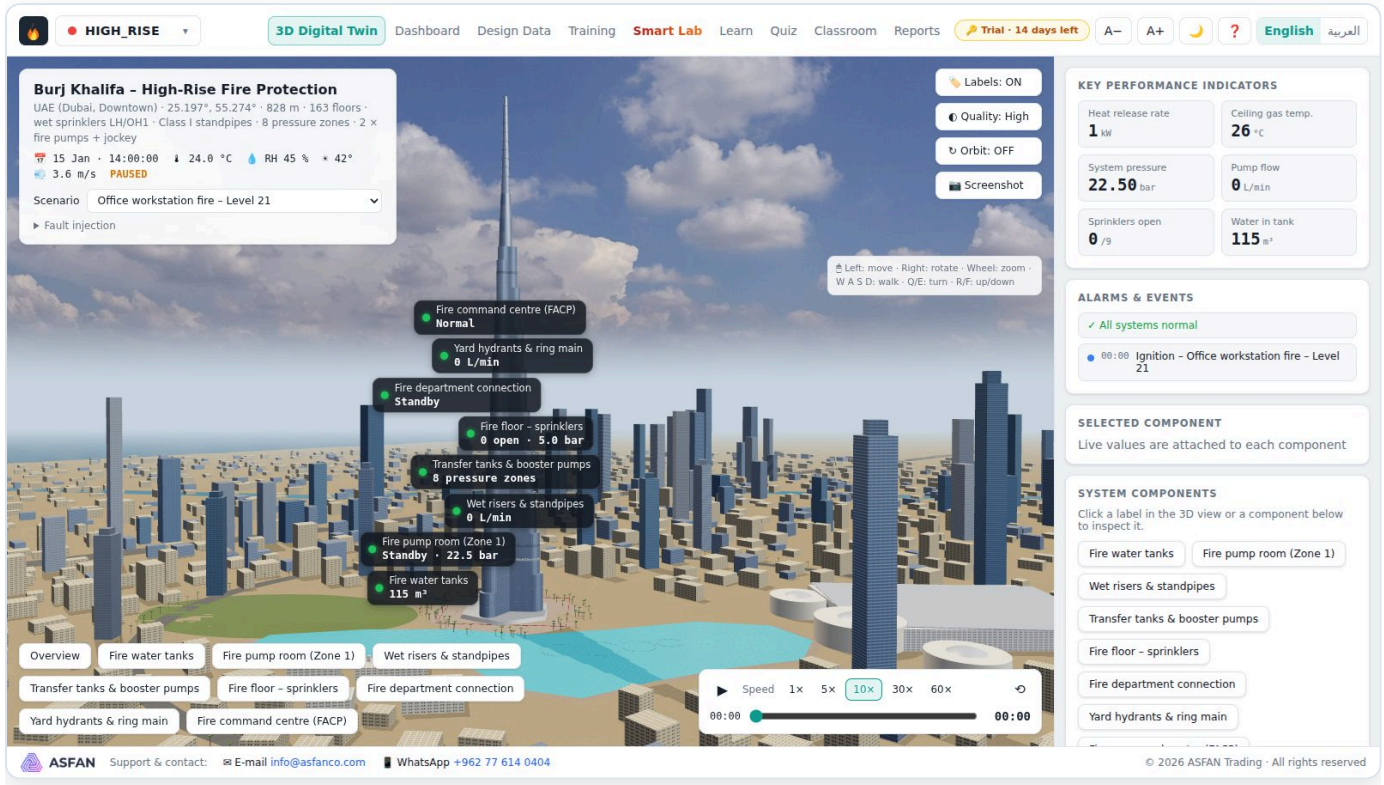
## Packages

A license can include the **full program** or only a **package**, for example *Sprinklers & pumps*, *Smart fire alarm*, *Special hazards* or *Basic awareness*, or a custom selection. The license window lists exactly which facilities, training scenes, Smart Lab modules and pages are included. Everything else is shown with a 🔒 lock; clicking it opens the license window so you can upgrade.

| Situation                    | What happens                                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------------|
| Subscription expires         | The program asks for a new key. Your data and results are kept.                                        |
| New computer                 | Unbound licenses can be activated on the new PC; bound licenses need a new key for the new Machine ID. |
| Supervisor (teacher) license | Sees, exports and clears the results of all students and the instructor dashboard.                     |

### **Do not change the computer date**

Setting the clock back is detected and blocks the program until the date is corrected.



Main screen: top bar, 3D view with live labels, information card, KPIs, alarms and components

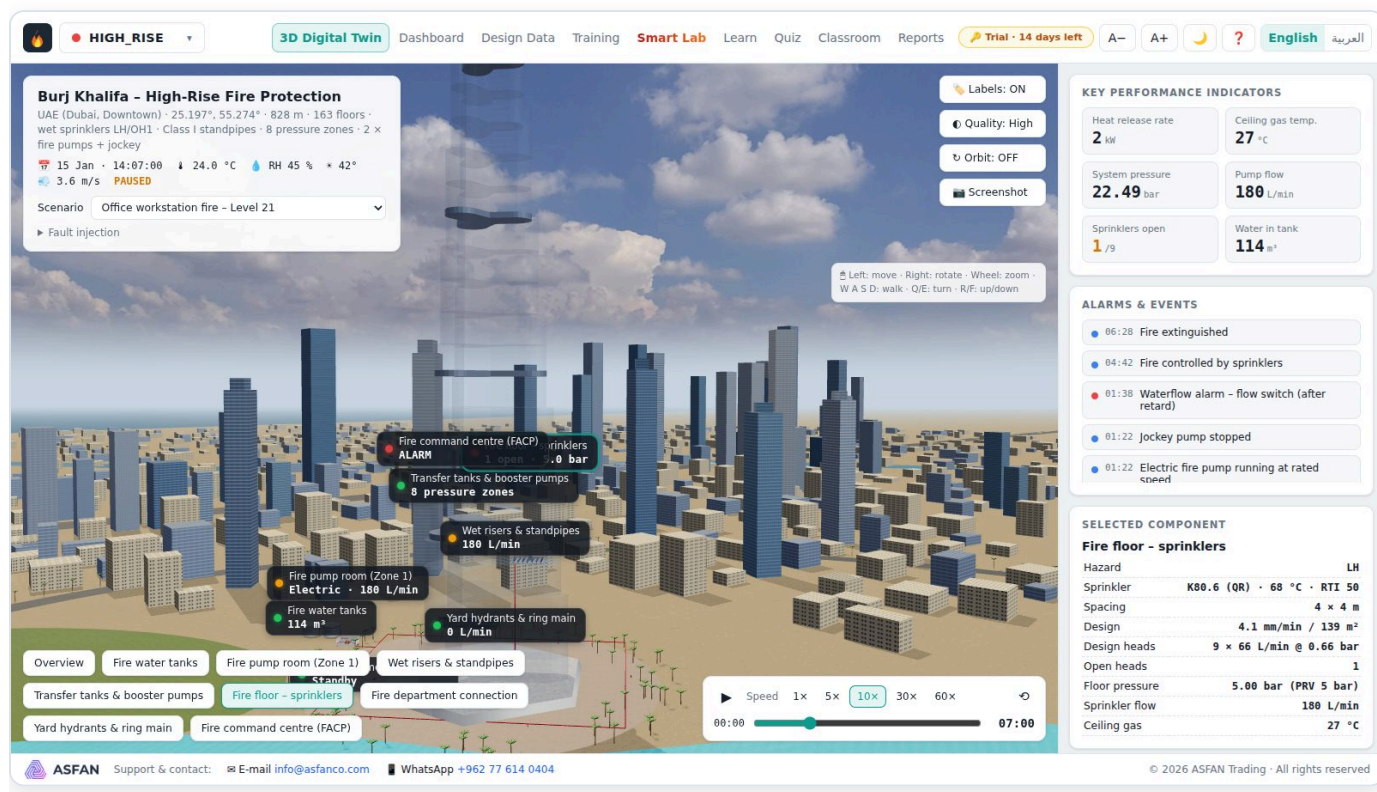
## Top bar

| Element           | Function                                                                                               |
|-------------------|--------------------------------------------------------------------------------------------------------|
| Facility selector | Choose one of the reference facilities or <b>Create facility manually</b> (your own building).         |
| Tabs              | 3D Digital Twin · Dashboard · Design Data · Training · Smart Lab · Learn · Quiz · Classroom · Reports. |
| 🔑 badge           | License status and activation.                                                                         |
| A- / A+           | Smaller / larger text.                                                                                 |
| 🌙 / ☀️            | Dark / light theme.                                                                                    |
| ?                 | Opens this user manual.                                                                                |
| English / العربية | Switch the whole program language (Arabic is right-to-left).                                           |

## Moving in 3D

| Action                  | Mouse / keyboard  |
|-------------------------|-------------------|
| Move (pan)              | Left button drag  |
| Rotate                  | Right button drag |
| Zoom                    | Mouse wheel       |
| Walk                    | W A S D           |
| Turn                    | Q / E             |
| Up / down               | R / F             |
| Play / pause simulation | Space             |

Buttons on the right of the 3D view: **Labels** on/off, **Quality** high/low (use *low* on slow laptops), **Orbit** auto-rotation, and **Screenshot** (also **Ctrl** + **Shift** + **S** ).

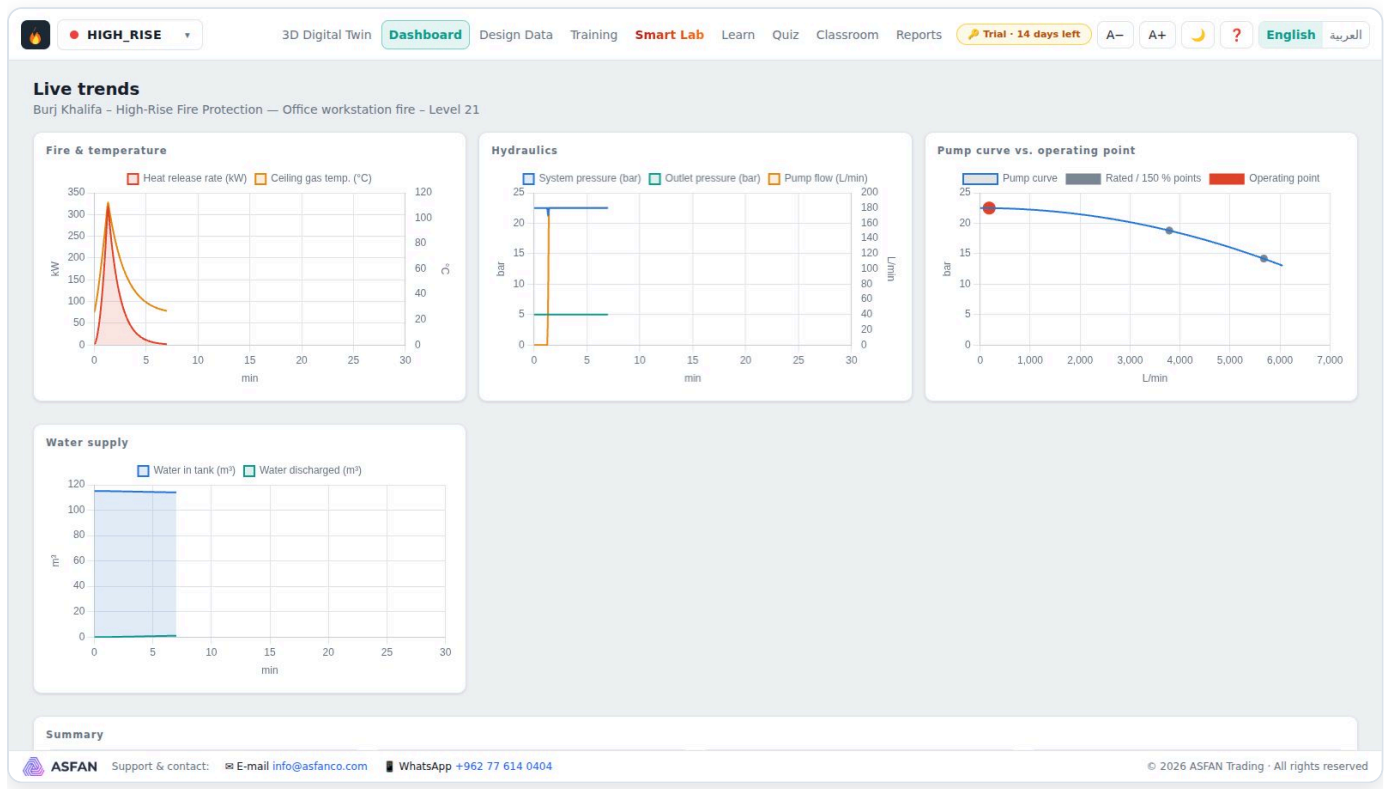


A fire scenario running: sprinklers operating, live pressures and alarms

- 1 Select a **facility** and a **scenario** (e.g. fire on a given floor).
- 2 Press ► to play. Use the speed buttons (1x–60x) and drag the time slider to jump.
- 3 Click any label or component to fly to it; its live values appear in **Selected component**.
- 4 Watch **KPIs** (heat release, gas temperature, pressure, flow, sprinklers open...) and the **Alarms & events** log.
- 5 Open **Fault injection** to add failures (power failure, closed valve, leak, pump failure...) and see how the system reacts.

### Create facility manually

Choose it from the facility selector to build your own building (city, occupancy, floors, dimensions, sprinkler K-factor, temperature rating, response type). The design and simulation are generated automatically.



Dashboard: fire, hydraulics, pump curve with moving operating point and water supply

Live trend charts for the running scenario: heat release and gas temperature, system pressure and flow, the fire-pump curve with the moving operating point, and tank/agent/foam supply. Charts follow the simulation time.

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### Design basis & calculations

Burj Khalifa - High-Rise Fire Protection — Office workstation fire - Level 21

**Edit design data - your own values**  
Change any value, then run the simulation on your data. Changed fields are highlighted; values are saved per facility and scenario.

**Sprinkler system (NFPA 13)**

|                                   |                                         |                                     |                                                     |                               |                               |
|-----------------------------------|-----------------------------------------|-------------------------------------|-----------------------------------------------------|-------------------------------|-------------------------------|
| Occupancy hazard<br>LH            | Sprinkler K-factor (L/min/vbar)<br>80.6 | Sprinkler spacing (square) (m)<br>4 | Response time index RTI ((m·s) <sup>½</sup> )<br>50 | Temperature rating (°C)<br>68 | Hose allowance (L/min)<br>379 |
| Water supply duration (min)<br>30 |                                         |                                     |                                                     |                               |                               |

**Piping & building**

|                            |                             |                                           |                                         |                         |                       |
|----------------------------|-----------------------------|-------------------------------------------|-----------------------------------------|-------------------------|-----------------------|
| Main pipe size (mm)<br>150 | Main pipe length (m)<br>140 | Fire floor elevation above pump (m)<br>88 | Floor PRV setting (0 = none) (bar)<br>5 | Ceiling height (m)<br>3 | Room length (m)<br>24 |
| Room width (m)<br>16       | Fire position x (m)<br>13   | Fire position y (m)<br>7                  |                                         |                         |                       |

**Fire scenario**

|                               |                                     |                                |
|-------------------------------|-------------------------------------|--------------------------------|
| Fire growth rate (t²)<br>fast | Peak heat release rate (kW)<br>3000 | Ambient temperature (°C)<br>24 |
|-------------------------------|-------------------------------------|--------------------------------|

**Pumps & water**

|                                           |                                           |                                      |                                    |
|-------------------------------------------|-------------------------------------------|--------------------------------------|------------------------------------|
| Fire pump rating (0 = auto) (gpm)<br>auto | Pump rated pressure (0 = auto) (bar)<br>0 | Number of duty pumps (0 = auto)<br>0 | Water storage (0 = auto) (m³)<br>0 |
|-------------------------------------------|-------------------------------------------|--------------------------------------|------------------------------------|

▶ Run simulation with my data
✓ Recalculate
↺ Reset to reference design
↓ Export JSON
↑ Import JSON

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Design Data: NFPA calculation steps, editable values and compliance checks

Shows every design calculation step (NFPA 13 density/area, K-factor, Hazen–Williams, pump selection, NFPA 2001 agent quantity, NFPA 11 foam...) in SI and US units. Change any value in the editor, press **Apply** or **Apply & run**, and the simulation runs on your design. The compliance list shows which NFPA rules pass or fail. **Reset** returns to the reference design.

**Training**  
Interactive 3D training scenes on real fire-protection equipment: explore every component, then perform the procedure step by step on live physics. Results are saved to the Classroom.

**New: Smart Systems Lab**  
Addressable panel, cause & effect, BMS integration, commissioning, predictive maintenance, incident command and certificates. [Open the Smart Lab →](#)

- Fire pump room walk-through & start sequence**  
NFPA 20 pump room: tank, suction, jockey / electric / diesel pumps, controllers, relief valve, test loop. Operate the demand and watch the pressure-switch sequence. [▶ Start](#)
- Annual fire pump flow test (NFPA 25)**  
Run the pump at churn, 100 % and 150 % of rated flow through the flow-meter loop, record net pressure and compare with the acceptance curve. [▶ Start](#)
- Dry-pipe valve: trip test & reset**  
Air holds the clapper closed; opening the inspector's test lets air escape, the valve trips and water must reach the test outlet within the NFPA 13 limit. Then restore the system step by step. [▶ Start](#)
- Floor control valve assembly: alarm & tamper tests**  
Zone control valve with tamper switch, waterflow switch with retard, gauge, test-and-drain with sight glass and PRV. Run the waterflow alarm test and the valve supervisory test. [▶ Start](#)
- Portable extinguishers: placement & PASS**  
Place 2-A-10-B-C extinguishers on a light-hazard office floor so every point is within 22.9 m (75 ft) travel distance (NFPA 10), then fight a fire using P-A-S-S. [▶ Start](#)
- Hose drill: landing valve & fire hose cabinet**  
Two-person team: open the cabinet, run out the 65 mm hose without kinks, couple and tug-test, fit the branch nozzle, open the landing valve slowly and attack the fire at its base. [▶ Start](#)
- Sprinkler types & response race**  
Compare pendent, upright, sidewall, concealed, recessed, ESFR and open heads, then light a t<sup>3</sup> fire under six bulbs of different temperature rating and RTI and watch which operates first. [▶ Start](#)
- Stairwell pressurization (NFPA 92)**  
Keep smoke out of the escape stair: set the supply fan so  $\Delta P \geq 12.5$  Pa, keep door-opening force  $\leq 133$  N, and see what happens when doors are opened.  $Q = 0.839 A \sqrt{\Delta P}$ . [▶ Start](#)
- FM-200 room: design calculator & discharge sequence**  
Size the agent live ( $W = V/S \times C/(100-C)$ ) with NOAEL/LOAEL warnings, then run the release sequence: cross-zoned detection, 30 s pre-discharge, HVAC shutdown, 10 s discharge and hold. [▶ Start](#)
- Installation mode: detectors & sprinklers**  
Lay out smoke detectors (0.7 × S rule) and light-hazard pendent sprinklers on a real office ceiling with a deep beam, supply diffusers, an exposed duct and a column — live coverage, NFPA 72 / NFPA 13 obstruction rules, and an installer's layout to correct. [▶ Start](#)

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The Training page with the ten interactive 3D scenes

| Scene                  | What you practise                                                                  |
|------------------------|------------------------------------------------------------------------------------|
| Fire pump room         | NFPA 20 components and the jockey → electric → diesel start sequence, manual stop. |
| Annual pump flow test  | Churn, 100 % and 150 % flow, acceptance curve (NFPA 25).                           |
| Dry-pipe valve         | Full trip test with water-delivery time and the reset procedure.                   |
| Floor control valve    | Waterflow alarm test (retard) and tamper / supervisory test.                       |
| Portable extinguishers | Placement to NFPA 10 travel distance and the P-A-S-S technique.                    |
| Hose drill             | Landing valve, couplings and two-person team operation.                            |
| Sprinkler types        | Pendent, upright, sidewall, concealed, ESFR... and a response race.                |
| Stair pressurization   | NFPA 92 pressure difference, door-opening force, relief damper.                    |
| FM-200 room            | NFPA 2001 calculator and the cross-zoned discharge sequence.                       |
| 3D installation mode   | Placing detectors and sprinklers on a real ceiling with obstruction rules.         |



A training scene (fire pump room) with the procedure panel on the right

## How a scene works

- 1 **Explore:** click any component (or its label) to read what it is and how it works.
- 2 Choose a **procedure**. The steps appear on the right; the current step is highlighted.
- 3 Perform each step by clicking components, moving sliders or answering questions. Wrong actions count as mistakes.
- 4 Use ×1 / ×5 / ×10 to speed up waiting times. At the end you get a score; results are saved to the Classroom.

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3D Digital Twin
Dashboard
Design Data
Training
Smart Lab
Learn
Quiz
Classroom
Reports
Trial - 14 days left
A-
A+
🌙
?
English
العربية

Labels: ON

Quality: High

Orbit: OFF

Screenshot

**Fire pump room walk-through & start sequence**

NFPA 20 pump room: tank, suction, jockey / electric / diesel pumps, controllers, relief valve, test loop. Operate the demand and watch the pressure-switch sequence.

🔍 **Explore**    Start sequence (NFPA 20)

- Check that the suction supply is open: click the supervised suction OS&Y valve.
- Identify the jockey (pressure-maintenance) pump.
- Open a small demand (= 2 %). The jockey must start at 8.61 bar.
- Increase the demand to = 60 % (sprinklers operating). The electric fire pump starts at 8.27 bar.
- Close the demand to 0 %. Does the electric fire pump stop by itself?
- With the demand closed, stop the electric pump: click its controller (STOP).
- Simulate a mains power failure: switch off "Mains power available".
- Open the demand again (= 60 %). The electric pump cannot start; the diesel cranks at 7.58 bar.
- Close the demand, restore mains power, then stop the diesel pump at its controller.

System demand (sprinklers / hoses open) 0 %

☒ Mains power available
 ☐ Small leak in the system

**LIVE VALUES**

System pressure 9.30 bar · 135 nci

X  
Exit

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# 10 Smart Systems Lab (advanced)

The **Smart Lab** tab trains engineers on smart (addressable) fire-alarm and life-safety systems. All ten modules share one live system: a fault planted in the panel appears in the building, the BMS, the monitoring centre and the commissioning records. A ✓ in the left menu marks completed modules.

The screenshot displays the 'Smart Systems Lab' interface for the 'Addressable Fire Alarm Panel Lab'. The top navigation bar includes '3D Digital Twin', 'Dashboard', 'Design Data', 'Training', 'Smart Lab' (active), 'Learn', 'Quiz', 'Classroom', 'Reports', and a 'Trial - 14 days left' badge. The left sidebar lists ten modules, with the first one, 'Addressable Fire Alarm Panel Lab', marked as completed with a green checkmark. The main content area is titled 'Addressable Fire Alarm Panel Lab' and describes the task: 'Operate a real-style addressable FACP connected to a 3-storey building; test every device in the field, cut and short the SLC loop to see Class A and isolators at work, program addresses with auto-learn, and diagnose hidden faults like a service engineer.' Below this, there are tabs for 'Live panel & building', 'Programming', 'Guided exercises', 'Inspector challenge', and 'Devices: install & wiring'. The 'Live panel & building' tab is active, showing a digital representation of the 'ASFAN FX-3000' fire alarm control panel. The panel's LCD displays 'SYSTEM NORMAL', 'Loop 1: 61 devices OK Class A', 'AC ON BATT 27.3V', and 'ACCESS LEVEL 1'. It features buttons for 'ACKNOWLEDGE', 'ALARM SILENCE', 'RESOUND', 'SYSTEM RESET', 'DRILL', 'LAMP TEST', and 'BUZZER'. To the right of the panel is a floor plan diagram of the 'Ground floor' showing various rooms and the SLC loop wiring. A legend at the bottom of the floor plan identifies symbols for SLC loop, NAC circuit, Isolator, Call point, Module, and Detector. Below the floor plan is a 'FIELD DEVICE' section with a prompt to 'Click a device on the plan to inspect and test it in the field.'

Module 1 – addressable fire alarm panel with the building plans and SLC loop

## 1 · Addressable panel lab

A realistic panel (LCD, LEDs, **ACKNOWLEDGE**, **ALARM SILENCE**, **SYSTEM RESET**, **DRILL**, lamp test and the access-level key switch). Click a device on the plan to test it (smoke, heat, call point, waterflow, valve). Use the toolbar to **cut** or **short** the loop, compare Class A and B, see isolators work, add devices and run **auto-learn** in the Programming tab. **Guided exercises** are detected automatically; the **Inspector challenge** hides faults for you to find. Silence and reset need access level 2 — turn the key.



## Module 3 – BMS graphics fed by the fire alarm gateway

Rules and diagrams ( $0.7 \times S$  spacing, wall distances, beam pockets, diffuser clearance, sprinkler spacing and the beam rule) and a button to open the 3D installation scene. A short quiz records your score.

Pre-functional checks (insulation test), device-by-device functional test of all devices (optionally with hidden installation defects), audibility survey with a sound-level meter, battery & power tests, sprinkler hydrostatic test and a printable NFPA 72 **Record of Completion**.

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**ADVANCED**

**Smart Systems Lab**  
Install, program, integrate, commission and maintain an addressable fire-alarm system.

1 **Addressable Fire Alarm Panel Lab**  
SLC loop, isolators, faults, progr...

2 **Cause & Effect Matrix**  
Program the smart logic & run fl...

3 **BMS & Building Integration**  
BACnet/Modbus, lift, voice evac...

4 **Installation Mode (3D)**  
Detector & sprinkler layout rules

5 **Commissioning & Acceptance Testing**  
Acceptance tests & record of co...

6 **Predictive Maintenance**  
Condition monitoring, ITM & wor...

7 **Incident Commander**  
Decisions under pressure, casca...

8 **Modern Detection & Suppression Technologies**  
ASD, beam, flame, LHD, mist, pr...

9 **Engineering Tools**  
Hydraulics, batteries, voltage dr...

10 **Learning Paths & Certificates**  
Levels, final exam, certificates

**Commissioning & Acceptance Testing**  
Take the installed system through acceptance exactly as a commissioning engineer does: pre-functional inspection and cable tests, a device-by-device functional test on the live panel, audibility and visibility survey, battery and power tests, the sprinkler hydrostatic test — and issue the NFPA 72 Record of Completion.

NFPA 72 §14.4 • T.14.4.3.2
NFPA 72 §7.5.6
NFPA 72 §18.4 • §18.5
NFPA 72 §10.6
NFPA 13 §29.2.1

1 **Pre-functional**
2 Device functional test
3 Audibility & visibility
4 Battery & power
5 Hydrostatic test
6 Record of Completion

Checklist items  
0/14

Flagged entries  
0

Installation defects recorded  
0

Loop class  
A

Programmed devices  
61

**PRE-FUNCTIONAL INSPECTION & CABLE TESTS**
NFPA 72 §14.4 • T.14.4.3.2

| Item / acceptance criterion                                                                                                                          | Measured                                  | Verdict                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <input type="radio"/> <b>Addressable devices, panel and surge protectors disconnected before insulation test</b><br><small>NFPA 72 §14.2.10</small>  | <input type="checkbox"/> Verified on site |                                                                                                                |
| <input type="radio"/> <b>SLC loop cable insulation resistance (conductors → earth)</b><br><small>≥ 1 MΩ @ 500 V DC • NFPA 72 T.14.4.3.2 (21)</small> | — MΩ                                      | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |
| <input type="radio"/> <b>NAC1 cable insulation resistance</b><br><small>≥ 1 MΩ @ 500 V DC • NFPA 72 T.14.4.3.2 (21)</small>                          | — MΩ                                      | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |
| <input type="radio"/> <b>NAC2 cable insulation resistance</b><br><small>≥ 1 MΩ @ 500 V DC • NFPA 72 T.14.4.3.2 (21)</small>                          | — MΩ                                      | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |
| <input type="radio"/> <b>SLC loop resistance, panel OUT → IN (continuity)</b><br><small>≤ 40 Ω • NFPA 72 T.14.4.3.2 (21)</small>                     | — Ω                                       | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |
| <input type="radio"/> <b>Earth-fault monitor: loop resistance to earth (panel reading)</b><br><small>≥ 50 kΩ • NFPA 72 §12.6</small>                 | — kΩ                                      | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |
| <input type="radio"/> <b>NAC1 end-of-line resistor</b><br><small>4.7 kΩ ± 5 % • NFPA 72 §12.6</small>                                                | — kΩ                                      | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |
| <input type="radio"/> <b>NAC2 end-of-line resistor</b><br><small>4.7 kΩ ± 5 % • NFPA 72 §12.6</small>                                                | — kΩ                                      | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |
| <input type="radio"/> <b>Loop scan: devices answering = devices programmed, no double address</b><br><small>= programmed • NFPA 72 §14.4.1</small>   | — dev.                                    | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |
| <input type="radio"/> <b>Max. smoke-detector spacing, smooth ceiling (corridor)</b><br><small>≤ 18.4 m • NFPA 72 §15.4.3.5</small>                   | — m                                       | <input type="button" value="Measure"/> <input type="button" value="Pass"/> <input type="button" value="Fail"/> |

**TEST INSTRUMENT**

**Why disconnect first?**  
A 500 V DC insulation test destroys the electronics of addressable detectors, modules and the panel. Megger the cables with devices removed from their bases and conductors lifted at the panel.

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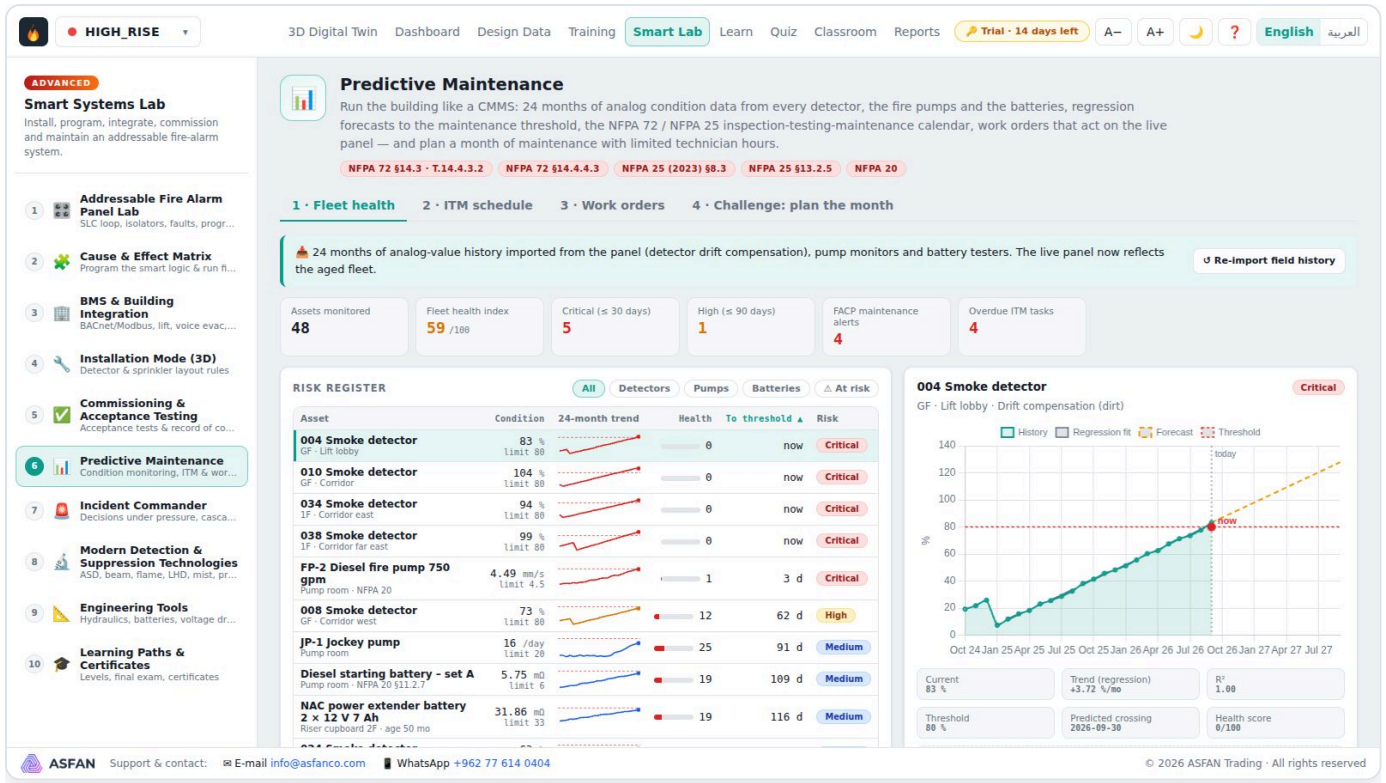
## Module 5 – commissioning: device tests, audibility, batteries, hydrostatic test, record

### 6 • Predictive maintenance

Condition data for detectors, pumps and batteries with trend forecasts, the NFPA 72 / NFPA 25 inspection-testing-maintenance calendar, work orders that really clean detectors on the panel, and a monthly planning challenge.

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Module 6 – fleet health, risk register and forecast

## 7 • Incident commander

Timed decisions in fires with cascading failures (night server-room fire, office-hours kitchen fire). Afterwards **Replay & debrief** shows your timeline, the best actions and their code basis.

## 8 • Modern detection & suppression

Simulators for aspirating smoke detection (VESDA), beam detectors, UV/IR flame detectors, linear heat cable, water mist and pre-action systems (non-, single- and double-interlock), plus a technology-selection challenge.



## 10 · Learning paths & certificates

Enter your name, follow a path (Technician, Engineer, Supervisor, Designer), complete the required modules, pass the final exam ( $\geq 80\%$ ) and print a certificate with a verification code. Supervisors see the instructor dashboard with each trainee's progress.

**HIGH\_RISE**

3D Digital Twin Dashboard Design Data Training **Smart Lab** Learn Quiz Classroom Reports Trial - 14 days left A- A+ 🌙 ? English العربية

**ADVANCED**  
**Smart Systems Lab**  
Install, program, integrate, commission and maintain an addressable fire-alarm system.

1 **Addressable Fire Alarm Panel Lab**  
SLC loop, isolators, faults, progr...

2 **Cause & Effect Matrix**  
Program the smart logic & run fi...

3 **BMS & Building Integration**  
BACnet/Modbus, lift, voice evac...

4 **Installation Mode (3D)**  
Detector & sprinkler layout rules

5 **Commissioning & Acceptance Testing**  
Acceptance tests & record of co...

6 **Predictive Maintenance**  
Condition monitoring, ITM & wor...

7 **Incident Commander**  
Decisions under pressure, casca...

8 **Modern Detection & Suppression Technologies**  
ASD, beam, flame, LHD, mist, pr...

9 **Engineering Tools**  
Hydraulics, batteries, voltage dr...

10 **Learning Paths & Certificates**  
Levels, final exam, certificates

**Learning Paths & Certificates**  
Four professional paths — Technician, Engineer, Supervisor, Designer. Complete the required labs, pass the final exam ( $\geq 80\%$ ) and print a certificate with a verification code that any employer can check offline in this program.  
NFPA 72 §10.5 Personnel qualifications NFPA 25 §4.4 NICET Level I-IV (reference)

🔗 Learning paths

📝 Final exam

🏆 Certificates & verification

👤 Instructor dashboard

Trainee name (printed on the certificate)

Group / company

 Save

**Fire Alarm Technician**  
Install, test and maintain addressable systems.

0%

☐ Panel lab0% / 60%

☐ Installation0% / 60%

☐ Commissioning0% / 60%

☐ Predictive maintenance0% / 60%

📝 Final exam

Complete the labs first

**Smart Systems Engineer**  
Program, integrate and commission complete smart fire systems.

0%

☐ Panel lab0% / 80%

☐ Cause & effect0% / 80%

☐ BMS integration0% / 60%

☐ Installation0% / 80%

☐ Commissioning0% / 70%

☐ Engineering tools0% / 60%

📝 Final exam

Complete the labs first

**Fire Safety Supervisor**  
Operate the building in emergencies and manage ITM & impairments.

0%

☐ Cause & effect0% / 70%

☐ BMS integration0% / 80%

**Fire Protection Designer**  
Design layouts, hydraulics, power and logic to code.

0%

☐ Cause & effect0% / 90%

☐ Installation0% / 90%

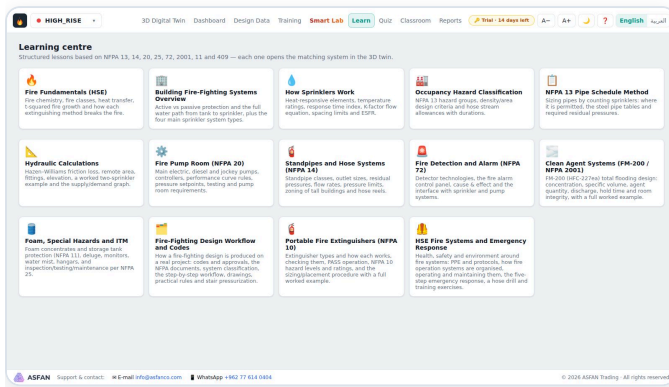
ASFAN Support & contact: [E-mail info@asfanco.com](mailto:info@asfanco.com) [WhatsApp +962 77 614 0404](https://www.whatsapp.com/channel/002916140404)

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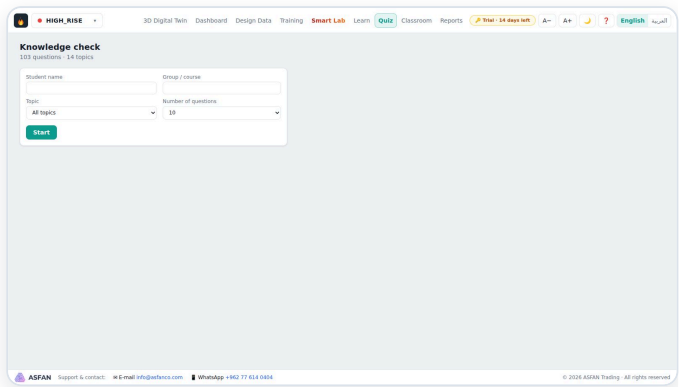
### Module 10 – learning paths, final exam and certificates

Fire Protection Digital Twin · User Manual

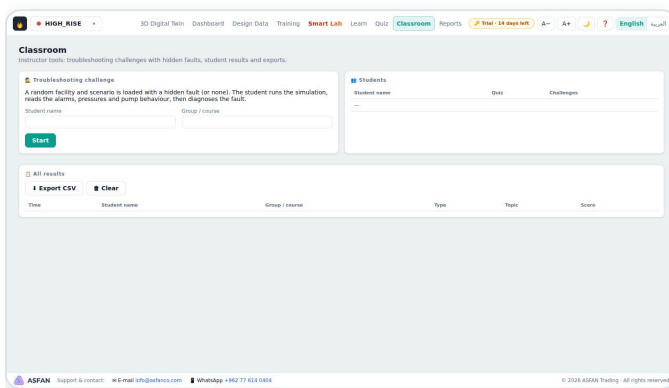
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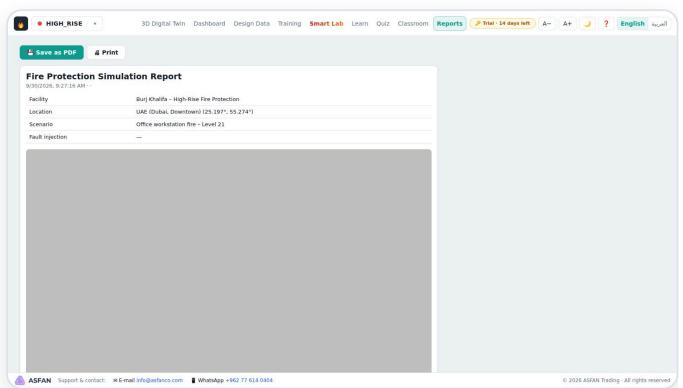
Learn – illustrated bilingual lessons



Quiz – questions with explanations



Classroom – challenges and results



Reports – printable simulation report

- **Learn:** illustrated lessons (fire basics, sprinklers, hazards, hydraulics, pumps, standpipes, detection & alarm, clean agents, foam, ITM).
- **Quiz:** choose a topic and number of questions; each answer shows an explanation.
- **Classroom:** enter the student name and group, start a **troubleshooting challenge** (hidden fault), and see results. A supervisor license can export and clear all results (CSV).
- **Reports:** **Save as PDF** or **Print** a full simulation report with screenshot, calculations and event log.

| Problem                                                                                                 | Solution                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| “Windows protected your PC” when installing                                                             | Click <b>More info</b> → <b>Run anyway</b> . The program is safe; the message appears because it is not signed with a paid certificate.             |
| 3D view is black, slow or flickering                                                                    | Switch <b>Quality</b> to <b>Low</b> ; update the graphics driver; on laptops choose the dedicated GPU for the program in Windows Graphics settings. |
| “License is bound to another computer”                                                                  | Send your current Machine ID to ASFAN to receive a key for this computer.                                                                           |
| “System clock rollback detected”                                                                        | Set the correct date and time in Windows and restart the program.                                                                                   |
| A tab or module shows  | It is not included in your package. Contact ASFAN to upgrade.                                                                                       |
| Text too small / large                                                                                  | Use <b>A–</b> / <b>A+</b> in the top bar.                                                                                                           |
| Results disappeared                                                                                     | Results are stored per Windows user; log in with the same Windows account.                                                                          |

## 13 Support & contact

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### **ASFAN Trading**

E-mail: **info@asfanco.com**

WhatsApp: **+962 77 614 0404**

When you contact support, please send: your name / organisation, your Machine ID (license window), the program version (Help → About) and a screenshot of the problem.